

Data Structure (2301CS301)

Semester	Credit	Contact Hours			Examination Scheme (Marks)				
					Theory		Practical		Total
		L	T	P	SEE	CE	SEE	CE	
3	5	3	-	4	40	30	20	10	100

Course Objective

To learn the basic types of Data Structures, their implementation, and their applications. To understand the importance of using suitable data structures for efficient programming. To develop the skill to identify appropriate data structures in problem-solving.

Prerequisite

Basic Programming

Course Outcome

At the end of this course, students will be able to:

1. understand the importance of data structure for efficient programming. (BL-2)
2. analyze the linear data structures and their applications. (BL-4)
3. implement the non-linear data structures and perform different operations on them. (BL-3)
4. describe the hashing functions and file structures. (BL-2)
5. differentiate the working of searching and sorting techniques. (BL-4)

Course Content

T: Teaching Hours, W: Weightage

Sr.	Topics	T	W
1	Introduction to Data Structure & Linear Data Structures: Array and Stack Data Structure: Data Management concepts, Data Types (Primitive & Non-primitive), Types of Data Structures (Linear & Non Linear Data Structures), Performance Analysis and Measurement (Time and space analysis of algorithms-Average, best and worst case analysis), Order Notations Array: Representation of Arrays, Sparse Matrix and its representation, Applications of Array Stack: Definitions & Concepts, Operations on Stack, Applications of Stack: Reverse a String, Conversion of Infix to Polish & Reverse Polish Notations, Evaluation of Polish Expressions, Recursion	9 hr	20 %
2	Linear Data Structures : Queue and Linked List Queue: Representation of Queue, Operations on Queue, Circular Queue, Double Ended Queue, Priority Queue, Applications of Queue Linked List: Representation, Operations & Applications - Singly, Circular & Doubly Linked List	10 hr	20 %
3	Nonlinear Data Structures : Tree and Graph Tree: Definitions and Concepts, Representation of Binary Tree, Conversion of General Tree to Binary Tree, Binary Tree Traversal (Preorder, Inorder & Postorder), Threaded Binary Tree, Binary Search Tree (BST), Balanced Trees: Height Balanced Tree (AVL Tree, 2-3 Tree, 2-3-4 Tree, Red-Black Tree), Weight Balanced Tree, Multiway Search Tree (B-Tree, B+ Tree), Trie, Applications of Tree Graph - Representation of Graph (Matrix Representation & Linked Representation), Graph Traversals : Breadth First Search (BFS) & Depth First Search (DFS), Cycle Detection in Graph, Spanning Trees, Minimum Spanning Tree, Prim's Algorithm, Kruskal's Algorithm, Finding the Shortest Path, Dijkstra's Algorithm	10 hr	20 %
4	Hashing and File Structures Hashing: Hash Table, Hash Functions, Collision-Resolution Techniques, Hashset, Hashmap File Structure: Concepts of fields, records and files, Sequential, Indexed and Relative/Random File Organization, Indexing structure for index files, hashing for direct files, Multi-Key file organization and access methods	8 hr	20 %

Sr.	Topics	T	W
5	Sorting and Searching Sorting: Internal & External Sorting, Bubble Sort, Selection Sort, Insertion Sort, Merge Sort, Quick Sort Searching: Linear Search, Binary Search	8 hr	20 %

Reference Books

1. An Introduction to Data Structures with Applications
| by Jean-Paul Tremblay & Paul G. Sorenson | Tata McGraw Hill
2. Data Structures using C & C++
| by Aaron M. Tanenbaum | PHI Learning
3. Fundamentals of Computer Algorithms
| by Ellis Horowitz, Sartaj Sahni, Sanguthever Rajasekaran | Universities Press (India) Private Limited | Ed. 2001
4. Fundamentals of Data Structures in C++
| by Sartaj Sahani | Ed. 2nd
5. Data and File Structures using C
| by Reema Thareja | Oxford University Press

Suggested List of Practicals / Tutorials

1. Hands-on practice to get familiar with basic programming concepts
2. Hands-on practice to get familiar with advanced programming concepts
3. Regular operations on 1-D Array Data Structure
4. Advanced operations on 1-D Array Data Structure
5. Operations on 2-D Array Data Structure
6. Implementation of Data Structure Stack
7. Solve real life problems using Stack
8. Conversion of Infix Expression to equivalent Postfix & Prefix Expression using Stack
9. Evaluation of Postfix & Prefix Expression using Stack
10. Implementation of Data Structure Queue & Circular Queue
11. Implementation of Data Structure Double Ended Queue & Priority Queue
12. Revision of the concept of Pointer & Dynamic Memory Allocation using C language
13. Implementation of Data Structure Singly Linked List
14. Basic Operations on Singly Linked List
15. Implementation of Stack & Queue using Singly Linked List
16. Perform some advanced operations on Singly Linked List - 1
17. Perform some advanced operations on Singly Linked List - 2
18. Implementation of Circular Linked List & Solving Problem using it
19. Implementation of Doubly Linked List & Solving Problem using it
20. Implementation of Binary Search Tree (BST)
21. Implementation of Height Balanced Tree
22. Perform advance operations on BST
23. Implementation of Data Structure Graph
24. Implementation of Data Structure Hash Table, Hashset & Hashmap
25. Implementation of Searching Techniques
26. Implementation of Bubble Sort & Selection Sort
27. Implementation of Insertion Sort & Merge Sort
28. Implementation of Quick Sort
29. Solving real life problem using sorting techniques
